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Proposal Review 2 : 2601631

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Agency Name:	National Science Foundation
Agency Tracking Number:	2601631
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Multigraded Homological Algebra and Geometry
Rating:	Multiple Rating: (Excellent/Very Good)

Review

Summary

Proposal Synopsis

The PI proposes a number of innovative projects dealing with syzygies and regularity in the multi-graded settings with a focus on stacky curves (in the spirit of Green's conjecture) and VG broader impacts include organization of conferences, mentoring and advocacy through various professional organizations.

Intellectual Merit

Strengths

The research projects presented are innovative extensions of well-established and central themes in commutative algebra and algebraic geometry like the use of syzygies and regularity to positivity properties of line bundles and linear series. The projects on stacky curves and multi-graded regularity are very well-motivated and the examples provided highlight the new challenges presented here. The strategy and techniques are well-reasoned and the previous work of the PI and her co-authors provides ample evidence of likely success. The third project on notions of regularity for the Cox category is more ambitious and is motivated by the recent advances on King's conjecture. If successful, this will be a breakthrough in the homological study of toric varieties. Even partial progress along the lines indicated in the proposal will be novel and interesting results in this theory.

Weaknesses

While the research project concerning the Cox category is an ambitious and exciting project, the strategy and techniques are not as extensively flushed-out as the other two projects.

Broader Impacts

Strengths

The PI has extensive experience organizing conferences and workshops in the area and is working on more in the coming years. The PI also has participated in professional organizations, leadership capacities and AMS committees and plans to continue doing so. The PI is also mentoring multiple students and expects to take on more. Based on these, the broader impacts of the proposals are very good and unique amongst the proposals I have reviewed thanks to the engagement with the community through professional organizations such as the AMS, the AWM, and the Spectra.

Weaknesses

The broader impacts would be even stronger with a more concrete plan for some of the workshops proposed. This could be achieved, for example, by expanding on the plans for the work Gems of Commutative Algebra or similar programs indicating the potential impacts.

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

Summary Statement

I rate this proposal Excellent/Very good and I have placed in the middle third of the proposals I reviewed. The proposal contains some very interesting projects with potential for substantial breakthroughs and discovering novel phenomena. The broader impacts are very good but could have been expanded upon further.

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